

Edit-by-Example: Adaptive Exemplar-Based Image Editing

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Abstract

Recent advances in diffusion-based image editing models have demonstrated remarkable success. However, these models primarily rely on high-quality textual prompts to guide image manipulation, creating a significant barrier for non-expert users. In this demonstration, we present an exemplar-based image editing framework named Edit-by-Example, which eliminates the reliance on textual prompts, requires only a single pair of before-and-after images to encapsulate the desired editing effect that can readily be applied on the user-provided query image without any model fine-tuning. Technically, our framework comprises two components: an Adaptive Editing Policy Module (AEPM) and a Generation Module (GM). The AEPM jointly analyzes cross-image relationships in exemplar pairs and query image content to derive optimal editing directions, while GM executes these policies through an off-the-shelf image editor with optional semantic alignment verification. We introduce EEdBench, a comprehensive benchmark for exemplar-based image editing containing 1,500 test cases across 15 categories. Experiments demonstrate that our framework outperforms existing prompt-free methods in editing direction accuracy (S-Visual) and fidelity (FID).

CCS Concepts

• Information systems → Multimedia content creation.

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1 Introduction

Recent years have witnessed remarkable advances in AIGC [5, 14, 15, 23], with particularly notable progress in the domain of

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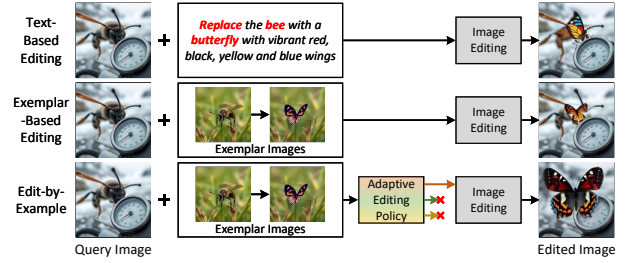


Figure 1: The text-based scheme takes a carefully crafted textual instruction to modify the query image. Exemplar-based approach can directly replicate the captured edit on the query image. We present Edit-by-Example, an exemplar-based image editing framework that adaptively generate optimal edit policy for better results.

image editing [4, 6, 20, 21]. Most existing approaches in this domain rely on textual prompts for image editing as shown in Figure 1 [12, 17, 22, 24]; however, this dependence introduces significant practical limitations. For example, achieving precise edits requires expertly engineered descriptions, excluding non-specialist users. Furthermore, practical scenarios requiring replication of specific editing effects on different query images (e.g., reproducing stylized outcomes from reference images) often demand iterative trial-and-error adjustments to textual prompts due to the language-vision gaps, compromising both efficiency and accessibility.

To eliminate the dependency on textual prompts, one intuitive solution is the exemplar-based image editing, which employs before-and-after image pairs as visual in-context instructions to encapsulate the desired editing effect, as illustrated in Figure 1. This idea is motivated by visual prompting [3] and developed by Analogist [8] and Reedit [18], which deploy Vision-Language Models (VLMs) to infer textual descriptions of edited query images. However, these approaches treat diverse editing types (e.g., object replacement vs. stylistic transfer) equivalently, assuming that VLMs can universally handle different exemplar and query images using one general instruction. To alleviate this limitation, we propose remolding the process through adaptive edit policy planning that prepares optimal inputs for off-the-shelf image editing models and applies post-hoc filtering when required. As such, the editing is dynamically tailored to diverse exemplar-query inputs.

Specifically, we present the Edit-by-Example, that enhances exemplar-based image editing through adaptive task categorization while preserving accessibility for non-expert users. The framework

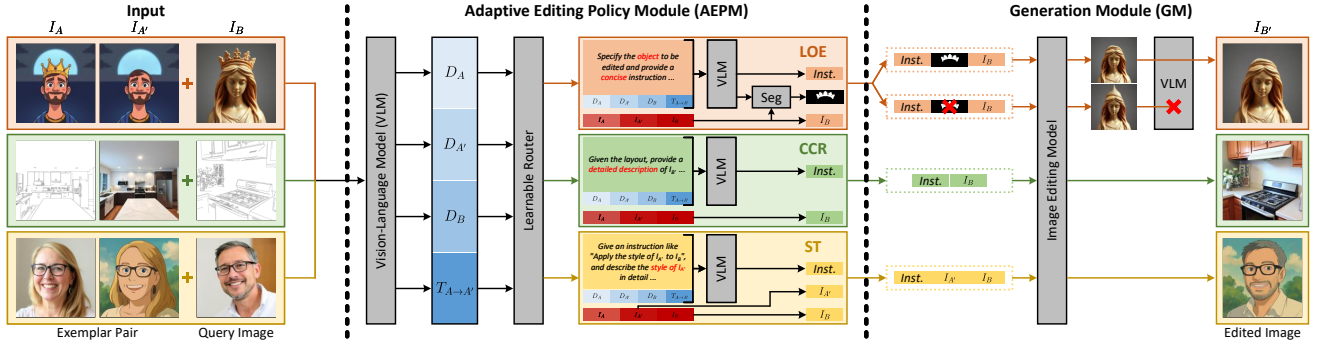


Figure 2: An overview of our Edit-by-Example framework. It processes diverse exemplar-query image inputs through the Adaptive Editing Policy Module (AEPM) to determine optimal editing policies, which guide the Generation Module (GM) in executing image modification.

begins by analyzing input exemplar to classify edits into three categories, each requiring a specialized workflow to optimize editing quality. Following this classification, category-specific processors prepare structured intermediate representations, including textual instructions, semantic masks, and conditioned images. These representations are then processed by an off-the-shelf image editor, with an optional post-selection refining object-level edits through semantic alignment checks to ensure better results. To demonstrate practical applicability, we have deployed our framework as a web application enabling users to achieve exemplar-based edits in an intuitive way. This implementation validates the framework’s effectiveness while showcasing its potential for real-world adoption.

2 Edit-by-Example

In this section we present our Edit-by-Example (shown in Figure 2) for image editing. It consists of two modules: an Adaptive Editing Policy Module (AEPM) and a Generation Module (GM). The AEPM analyzes input exemplar pairs and query images to derive category-specific policies, generating optimized intermediate data for GM, which produces final edited images.

Problem Formulation. Given before-and-after images $\langle I_A, I_{A'} \rangle$ as an exemplar, the goal of Edit-by-Example is replicating the transformation from I_A to $I_{A'}$ on the query image I_B and obtaining the edited image $I_{B'}$. We denote such process as $I_{B'} = \mathcal{F}(I_B | \langle I_A, I_{A'} \rangle)$.

Adaptive Editing Policy Module. The AEPM is mainly designed to adaptively choose the way to prepare the intermediate data which are optimized for the generation of GM. Given user inputs I_A , $I_{A'}$, and I_B , it first describes in detail the visual contents and styles of them (denoted as D_A , $D_{A'}$, and D_B), and summarizes the transformation from I_A to $I_{A'}$ ($T_{A \rightarrow A'}$) via comparing the difference between two images in the exemplar. Then, a learnable router is employed to classify the edit into three categories: Localized Object-aware Editing (LOE) for precise manipulations such as object insertion and deletion, Cross-modal Content Realization (CCR) for synthesizing images from sketches, depth maps, or scribbles, and Style Transfer (ST) for artistic conversions like photorealistic-to-3D-art transformations. Conditioning on input images ($I_A/I_{A'}/I_B$) and their VLM-generated descriptions ($D_A/D_{A'}/D_B/T_{A \rightarrow A'}$), we implement specialized processing pipelines tailored to each edit category. The VLM uses a prompt that encourages the model to engage in multi-round reasoning and self-correction. For LOE where only object-relevant regions require modification, we enforce concise

Table 1: Performance Comparisons on EEdBench.

Methods	Analogist [8]	PromptDiff. [19]	InstaManip [11]	Ours
S-Visual ($\uparrow$)	0.1469	0.1405	0.2234	0.2382
FID ($\downarrow$)	37.18	92.92	36.26	27.79

textual instructions (e.g., "Add a crown") to suppress VLM hallucinations while generating segmentation masks, jointly constraining editable regions to prevent over-modification. CCR addresses sparse inputs like sketches/depth maps by prompting VLMs to extrapolate contextual details, enriching I_B ’s edit instructions with material textures and lighting conditions absent in original layouts. To better preserve the style in $I_{A'}$, which is hard to be described by text, ST passes $I_{A'}$ to GM in addition to I_B .

Generation Module. Leveraging outputs from AEPM, GM executes the edit on I_B using an off-the-shelf image editing model. Additionally, to address segmentation mask ambiguities in LOE, we generate paired outputs (with/without mask as input) and employ a VLM to select the result that better meet the desired edit.

Implementation. We employ Qwen2.5-VL [2] as the VLM and ACE [9] as the image editing model. The Bert-based [7] learnable router is trained on manually categorized editing tasks from AnyEdit [20]. And we use the Grounding DINO [13]/SAM [10] to segment objects, and Gradio [1] to build web application.

Experimental Results. To verify the merit of our proposal, we introduce EEdBench, an exemplar-based image editing benchmark comprising 15 editing categories, each containing 100 annotated test cases. Following [8, 16], we use S-Visual for editing direction which quantifies the alignment between the visual change in $\langle I_A, I_{A'} \rangle$ and $\langle I_B, I_{B'} \rangle$, and the FID to measure the visual fidelity between the prediction $I_{B'}$ and ground truth of editing $I_{B'}^{gt}$. Table 1 shows the comparisons with baselines including Analogist [8], PromptDiffusion [19], and InstaManip [11]. The results validate our proposal.

3 Application and User Experience

We deploy Edit-by-Example as a user-friendly web application, enabling prompt-free editing through exemplar inputs. The backend runs on an Nvidia A800 GPU, and the frontend operates via a browser interface on PC or laptop. Users upload/select a before-and-after image pair, followed by the query image. The system processes the input and returns the edited result, requiring no prompt engineering or model customization. The edited result can also be saved as a new query image for subsequent rounds of editing.

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